

CLAIM-ARGUMENT-EVIDENCE (CAE) GOAL STRUCTURING TREE FOR EMBODIED RELEASE

Autonomous Tactile Workstation: High-Level Safety Claim → Evidence Grounding → Defeater Analysis → Conditional Release

TOP OPERATIONAL SAFETY CLAIM (C₁)

Falsifiable Physical Boundary

"Manipulator never exerts normal force $F_{\text{norm}} > 50 \text{ N}$ or shear force $F_{\text{shear}} > 20 \text{ N}$ on unexpected obstacle or human operator across any operational state, with emergency stopping time $t_{\text{stop}} \leq 40 \text{ ms}$."

CONTEXT: OPERATIONAL DESIGN DOMAIN (X_ODD)

- Reach $r \leq 1.2 \text{ m}$ · Max velocity $v_{\text{max}} = 0.40 \text{ m/s}$
- Payload mass $m \leq 2.0 \text{ kg}$ (nominal) · Dry friction $\mu \geq 0.35$
- Temp: $15^{\circ}\text{C} - 40^{\circ}\text{C}$ · Light: $300 - 2000 \text{ lx}$ · 1 co-present worker

SUB-CLAIM C_{1,1}: Tactile Reflex

Primary Soft Tactile Slip Reflex

Optical-tactile fingertips detect contact shear & trip safety reflex within 12 ms; normal force arrested $< 14 \text{ N}$.

ARGUMENT A₁ (Tactile Shear Warrant)

Elastomer marker displacement tracks surface shear under Coulomb friction assumption ($\mu = 0.85$).
Assumes: Clean, dry contact interface

× DEFEATER D₁: OIL CONTAMINATION (BROKEN LINK)

Machining fluid drops friction: $\mu = 0.85 \rightarrow 0.12$.
Workpiece slides freely across silicone; zero marker strain observed. Primary tactile reflex fails silently!
Result: Unmonitored rigid collision trajectory

EVIDENCE E₁

Tactile Metrology

- Calibrated pad stiffness $k = 4.0 \text{ N/mm}$
- P99.9 perception latency: 12 ms
- 10^5 HIL dropouts: 0 uncontained trips
- Trace: LOG-TAC-2026-0814

SUB-CLAIM C_{1,2}: Current Observer

Secondary Torque Tripwire

Joint servo drives observe motor current discrepancy $\Delta t > 3.5 \text{ N-m}$ to command friction brakes independently.

ARGUMENT A₂ (Current Tripwire Warrant)

Motor torque observer trips at $\Delta t = 3.5 \text{ N-m}$ ($F_{\text{trip}} = 35 \text{ N}$). Mechanical brakes halt moving arm.
Unmitigated: Peak force reaches $88 \text{ N} > 50 \text{ N}!$

△ COMPENSATORY CONDITIONAL BRANCH

- Derate payload mass: $m \leq 1.5 \text{ kg}$ ($m_{\text{eff}} = 4.0 \text{ kg}$)
- Add optical surface-dryness station at feeder
- Tighten torque tripwire threshold to $F_{\text{trip}} = 18 \text{ N}$
Bound Restored: $F_{\text{peak}} = 38 \text{ N} \leq 50 \text{ N}$ ($\int F dt = 1.1 \text{ N-s}$)

EVIDENCE E₂

Impact Mechanics Proof

- Dynamometer $a_{\text{brake}} = 8.0 \text{ m/s}^2$
- Impulse integral: $\int F dt = 1.1 \text{ N-s}$
- Max contact force: $F_{\text{peak}} = 38 \text{ N}$
- Trace: DYN-IMP-2026-0922

EVIDENCE E₃

Optical Dryness Check

- Feeder strobe specular detector
- Threshold: Fluid sheen $< 2\%$ area
- Automatic rejection of oily stock
- Trace: OPT-DRY-2026-0301

SUB-CLAIM C_{1,3}: Deterministic Gate

Hard Real-Time Enforcer (Ch 12)

1000 Hz safety loop isolates inverter gate signals within 1 ms; real-time bus latency bounded $P99.99 \leq 1.8 \text{ ms}$.

ARGUMENT A₃ (Architectural Warrant)

Dedicated MCU hardware interlock line cannot be preempted, starved, or bypassed by MPU vision tasks.
Independent permission path (Chapter 12)

✓ VERIFIED ARCHITECTURAL INDEPENDENCE

- Inverter gate power cut latency: $t_{\text{gate}} = 1.0 \text{ ms}$
- Holding brake engage latency: $t_{\text{brake}} = 18.0 \text{ ms}$
- Watchdog hardware dead-man timeout: 50 ms
Strict forward invariance guaranteed in hardware

EVIDENCE E₄

MCU Execution Traces

- Logic analyzer trace: $1 \text{ kHz} \pm 5 \mu\text{s}$
- Current filter delay $t_{\text{sense}} = 6 \text{ ms}$
- SPI DMA bus load: 0 overruns
- Trace: LOG-MCU-2026-1105

ADJUDICATION RESOLUTION & VERDICT CONTRACT

× UNCONDITIONAL OPERATE

STATUS: REFUSED / REJECTED

Defeater D₁ invalidates unconstrained slip reflex.
Peak contact force reaches 88 N ($> 50 \text{ N}$ limit).

✓ OPERATE WITH CONDITIONS

STATUS: APPROVED & SIGNED

- Payload $\leq 1.5 \text{ kg}$ · 2. Feeder optical check
- $F_{\text{trip}} = 18 \text{ N}$ tripwire $\rightarrow F_{\text{peak}} = 38 \text{ N} \leq 50 \text{ N}$

△ STANDING CONDITIONS & EXPIRY

LIFECYCLE CONTRACT

- Sensor recalibration every 500 operating hours
- Automatic invalidation upon oil mist detection